MRV System — Database Schema

Entity Relationship Diagram

```
┌──────────────────┐        ┌──────────────────┐        ┌──────────────────┐
│  organizations   │        │      users       │        │    countries     │
├──────────────────┤        ├──────────────────┤        ├──────────────────┤
│ id          (PK) │◄──┐    │ id          (PK) │        │ id          (PK) │
│ name             │   │    │ cognito_sub (UQ) │        │ name             │
│ type (enum)      │   │    │ email       (UQ) │        │ iso_code    (UQ) │
│ status           │   ├────│ organization_id  │   ┌────│ organization_id  │
│ created_at       │   │    │ role             │   │    │ auditor_org_id   │──┐
│ updated_at       │   │    │ status           │   │    │ created_at       │  │
└──────────────────┘   │    │ first_name       │   │    │ updated_at       │  │
        ▲              │    │ last_name        │   │    └──────────────────┘  │
        │              │    │ last_login_at    │   │            ▲             │
        │              │    │ created_at       │   │            │             │
        │              │    │ updated_at       │   │            │             │
        └──────────────┴────┘                    │            │             │
                       │                         │            │             │
        ┌──────────────┘                         │            │             │
        │    organizations                       │            │             │
        │    (type=COUNTRY) ─────────────────────┘            │             │
        │    (type=AUDITOR) ──────────────────────────────────┘             │
        │                                                                   │
        └───────────────────────────────────────────────────────────────────┘

┌────────────────────┐        ┌────────────────────┐
│  report_templates  │        │  template_sections │
├────────────────────┤        ├────────────────────┤
│ id            (PK) │◄───────│ template_id    (FK) │
│ name               │        │ id             (PK) │
│ description        │        │ title               │
│ version            │        │ instructions  (jsonb)│
│ status (enum)      │        │ editor_type   (enum) │
│ template_group_id  │        │ editor_config (jsonb) │
│ created_by      (FK) │──►users │ sample_data  (jsonb) │
│ published_at       │        │ sort_order          │
│ created_at         │        │ created_at          │
│ updated_at         │        │ updated_at          │
└────────────────────┘        └────────────────────┘

┌────────────────────┐        ┌────────────────────┐
│      reports       │        │   report_sections  │
├────────────────────┤        ├────────────────────┤
│ id            (PK) │◄───────│ report_id      (FK) │
│ template_id   (FK) │──►report_templates │ id             (PK) │
│ template_version   │        │ template_section_id (FK)│──►template_sections
│ country_id    (FK) │──►countries │ content       (jsonb) │
│ status (enum)      │        │ is_complete         │
│ current_revision   │        │ last_edited_by  (FK) │──►users
│ created_by    (FK) │──►users │ last_edited_at      │
│ submitted_at       │        │ version (optimistic) │
│ completed_at       │        │ created_at          │
│ created_at         │        │ updated_at          │
│ updated_at         │        └────────────────────┘
└────────────────────┘
        │
        │  1:N
        ▼
┌────────────────────┐        ┌────────────────────┐
│  report_revisions  │        │    change_records  │
├────────────────────┤        ├────────────────────┤
```

```
│ id            (PK)   │◄──┐   │ id              (PK) │
│ report_id     (FK)   │   │   │ report_id       (FK) │──►reports
│ revision_number      │   │   │ section_id      (FK) │──►report_sections
│ content_snapshot(jsonb)│  ├──│ from_revision_id(FK) │
│ submitted_by  (FK)   │──►users  │ to_revision_id  (FK) │
│ submitted_to         │   │   │ previous_content(jsonb)│
│ submitted_at         │   │   │ new_content    (jsonb) │
│ created_at           │   │   │ created_at           │
└──────────────────────┘   │   └──────────────────────┘
                           │
                           │
┌──────────────────────┐   │
│      comments        │   │
├──────────────────────┤   │
│ id            (PK)   │   │
│ report_id     (FK)   │──►reports
│ section_id    (FK)   │──►report_sections
│ revision_id   (FK)   │──┘
│ parent_comment_id(FK)│──►comments (self-ref, threads)
│ author_id     (FK)   │──►users
│ content              │
│ status (enum)        │
│ resolved_by   (FK)   │──►users
│ resolved_at          │
│ created_at           │
│ updated_at           │
└──────────────────────┘

┌──────────────────────┐        ┌──────────────────────┐
│    notifications     │        │     audit_log        │
├──────────────────────┤        ├──────────────────────┤
│ id            (PK)   │        │ id              (PK) │
│ user_id       (FK)   │──►users  │ user_id        (FK) │──►users
│ type (enum)          │        │ action               │
│ title                │        │ entity_type          │
│ message              │        │ entity_id            │
│ link                 │        │ metadata      (jsonb) │
│ is_read              │        │ created_at           │
│ created_at           │        └──────────────────────┘
└──────────────────────┘
```

Enums

org_type

Value	Description
`WORLD_BANK`	FMT / World Bank organization
`COUNTRY`	Participating country organization
`AUDITOR`	Auditing entity organization

user_status

Value	Description
`ACTIVE`	User can log in and access the system
`INACTIVE`	User is deactivated, cannot log in

user_role

Value	Description
`ADMIN`	Organization administrator
`MEMBER`	Regular organization member

template_status

Value	Description
`DRAFT`	Template is being edited, not visible to countries
`PUBLISHED`	Template is active and can be used by countries
`ARCHIVED`	Template is retired, no new reports can use it

editor_type

Value	Description
`RICH_TEXT`	TipTap rich text editor (headings, lists, paragraphs)
`TABLE`	Structured table editor (columns defined in editor_config)

report_status

Value	Description	Editable?
`DRAFT`	Being filled by country user	Yes
`SUBMITTED_TO_FMT`	Awaiting FMT pickup	No
`FMT_REVIEW_IN_PROGRESS`	FMT is actively reviewing	No
`FMT_REVISION_REQUESTED`	FMT returned with comments	Yes
`FMT_APPROVED`	FMT approved, ready for auditor	No
`SUBMITTED_TO_AUDITOR`	Forwarded to assigned auditor	No
`AUDITOR_REVIEW_IN_PROGRESS`	Auditor is actively reviewing	No
`AUDITOR_REVISION_REQUESTED`	Auditor returned with comments	Yes
`COMPLETED`	Final, report is locked	No

comment_status

Value	Description
`OPEN`	Comment needs to be addressed
`RESOLVED`	Comment has been addressed

notification_type

Value	Description
`REPORT_SUBMITTED`	A report was submitted for review
`REVISION_REQUESTED`	A reviewer requested changes
`REPORT_RESUBMITTED`	A revised report was resubmitted

Value	Description
`REPORT_COMPLETED`	An auditor marked a report complete
`NEW_TEMPLATE_VERSION`	A new template version was published

Table Definitions

organizations

Column	Type	Nullable	Default	Constraints
id	uuid	No	gen_random_uuid()	PRIMARY KEY
name	varchar(255)	No	-	-
type	org_type	No	-	-
status	user_status	No	'ACTIVE'	-
created_at	timestamptz	No	now()	-
updated_at	timestamptz	No	now()	-

users

Column	Type	Nullable	Default	Constraints
id	uuid	No	gen_random_uuid()	PRIMARY KEY
cognito_sub	varchar(255)	No	-	UNIQUE
email	varchar(255)	No	-	UNIQUE
first_name	varchar(255)	No	-	-
last_name	varchar(255)	No	-	-
organization_id	uuid	No	-	FK → organizations(id)
role	user_role	No	'MEMBER'	-
status	user_status	No	'ACTIVE'	-
last_login_at	timestamptz	Yes	NULL	-
created_at	timestamptz	No	now()	-
updated_at	timestamptz	No	now()	-

Indexes: - `idx_users_cognito_sub` on (cognito_sub) — auth lookup - `idx_users_email` on (email) — unique lookup - `idx_users_organization_id` on (organization_id) — list users by org

countries

Column	Type	Nullable	Default	Constraints
id	uuid	No	gen_random_uuid()	PRIMARY KEY
name	varchar(255)	No	-	-
iso_code	varchar(3)	No	-	UNIQUE

Column	Type	Nullable	Default	Constraints
organization_id	uuid	No	-	FK → organizations(id), UNIQUE
auditor_organization_id	uuid	Yes	NULL	FK → organizations(id)
created_at	timestamptz	No	now()	-
updated_at	timestamptz	No	now()	-

Indexes: - `idx_countries_organization_id` on (organization_id) - `idx_countries_auditor_organization_id` on (auditor_organization_id)

Constraints: - auditor_organization_id must reference an organization with type = 'AUDITOR' (enforced at application level)

report_templates

Column	Type	Nullable	Default	Constraints
id	uuid	No	gen_random_uuid()	PRIMARY KEY
name	varchar(255)	No	-	-
description	text	Yes	NULL	-
version	integer	No	1	-
status	template_status	No	'DRAFT'	-
template_group_id	uuid	No	gen_random_uuid()	Links all versions of same template
created_by	uuid	No	-	FK → users(id)
published_at	timestamptz	Yes	NULL	-
created_at	timestamptz	No	now()	-
updated_at	timestamptz	No	now()	-

Indexes: - `idx_templates_group_status_version` on (template_group_id, status, version DESC) — find latest published - `idx_templates_status` on (status) — list by status

Versioning: Each new version is a separate row. All versions of the same logical template share `template_group_id`. To find the latest published: `WHERE template_group_id = ? AND status = 'PUBLISHED' ORDER BY version DESC LIMIT 1`.

template_sections

Column	Type	Nullable	Default	Constraints
id	uuid	No	gen_random_uuid()	PRIMARY KEY
template_id	uuid	No	-	FK → report_templates(id) ON DELETE CASCADE
title	varchar(255)	No	-	-
instructions	jsonb	No	'{}'	TipTap JSON document
editor_type	editor_type	No	'RICH_TEXT'	-
editor_config	jsonb	No	'{}'	Configuration for the editor
sample_data	jsonb	Yes	NULL	Pre-filled content

Column	Type	Nullable	Default	Constraints
sort_order	integer	No	-	-
created_at	timestamptz	No	now()	-
updated_at	timestamptz	No	now()	-

Indexes: - `idx_template_sections_template_order` on (template_id, sort_order)

editor_config for TABLE type:

```
{
  "columns": [
    { "key": "year", "label": "Year", "type": "number", "required": true },
    { "key": "value", "label": "Emission Value (tCO2e)", "type": "number", "required": true },
    { "key": "source", "label": "Data Source", "type": "text", "required": false }
  ]
}
```

editor_config for RICH_TEXT type:

```
{
  "features": ["headings", "bold", "italic", "bulletList", "orderedList", "table"]
}
```

reports

Column	Type	Nullable	Default	Constraints
id	uuid	No	gen_random_uuid()	PRIMARY KEY
template_id	uuid	No	-	FK → report_templates(id)
template_version	integer	No	-	Snapshot of template version at creation
country_id	uuid	No	-	FK → countries(id)
status	report_status	No	'DRAFT'	-
current_revision	integer	No	0	Incremented on each submission
created_by	uuid	No	-	FK → users(id)
submitted_at	timestamptz	Yes	NULL	Last submission timestamp
completed_at	timestamptz	Yes	NULL	When auditor marked complete
created_at	timestamptz	No	now()	-
updated_at	timestamptz	No	now()	-

Indexes: - `idx_reports_country_status` on (country_id, status) — country user's report list - `idx_reports_status` on (status) — FMT/auditor review queues - `idx_reports_template_id` on (template_id) — reports using a template

report_sections

Column	Type	Nullable	Default	Constraints
id	uuid	No	gen_random_uuid()	PRIMARY KEY
report_id	uuid	No	-	FK → reports(id) ON DELETE CASCADE

Column	Type	Nullable	Default	Constraints
template_section_id	uuid	No	-	FK → template_sections(id)
content	jsonb	No	'{}'	TipTap JSON or table data
is_complete	boolean	No	false	Whether section has content
last_edited_by	uuid	Yes	NULL	FK → users(id)
last_edited_at	timestamptz	Yes	NULL	-
version	integer	No	1	For optimistic locking
created_at	timestamptz	No	now()	-
updated_at	timestamptz	No	now()	-

Indexes: - `idx_report_sections_report_id` on (report_id)

Optimistic locking: Client sends `If-Match: <version>` header. Server checks `WHERE version = <expected>`. On match, increments version and saves. On mismatch, returns 409 Conflict.

Content for TABLE type:

```
{
  "rows": [
    { "year": 2024, "value": 1500.5, "source": "National Inventory" },
    { "year": 2025, "value": 1420.3, "source": "Satellite Data" }
  ]
}
```

Content for RICH_TEXT type:

```
{
  "type": "doc",
  "content": [
    { "type": "paragraph", "content": [{ "type": "text", "text": "..." }] }
  ]
}
```

report_revisions

Column	Type	Nullable	Default	Constraints
id	uuid	No	gen_random_uuid()	PRIMARY KEY
report_id	uuid	No	-	FK → reports(id) ON DELETE CASCADE
revision_number	integer	No	-	Matches reports.current_revision at time of creation
content_snapshot	jsonb	No	-	Full copy of all section content at submission
submitted_by	uuid	No	-	FK → users(id)
submitted_to	varchar(20)	No	-	'fmt' or 'auditor'
submitted_at	timestamptz	No	now()	-
created_at	timestamptz	No	now()	-

Indexes: - `idx_revisions_report_number` on (report_id, revision_number)

content_snapshot format:

```
{
  "sections": {
    "<section_id>": {
      "title": "Section Title",
      "editor_type": "RICH_TEXT",
      "content": { ... },
      "is_complete": true
    }
  }
}
```

comments

Column	Type	Nullable	Default	Constraints
id	uuid	No	gen_random_uuid()	PRIMARY KEY
report_id	uuid	No	-	FK → reports(id) ON DELETE CASCADE
section_id	uuid	No	-	FK → report_sections(id)
revision_id	uuid	No	-	FK → report_revisions(id)
parent_comment_id	uuid	Yes	NULL	FK → comments(id) — for threading
author_id	uuid	No	-	FK → users(id)
content	text	No	-	-
status	comment_status	No	'OPEN'	-
resolved_by	uuid	Yes	NULL	FK → users(id)
resolved_at	timestamptz	Yes	NULL	-
created_at	timestamptz	No	now()	-
updated_at	timestamptz	No	now()	-

Indexes: - `idx_comments_report_section` on (report_id, section_id) — load comments per section - `idx_comments_report_status` on (report_id, status) — count open comments

change_records

Column	Type	Nullable	Default	Constraints
id	uuid	No	gen_random_uuid()	PRIMARY KEY
report_id	uuid	No	-	FK → reports(id) ON DELETE CASCADE
section_id	uuid	No	-	FK → report_sections(id)
from_revision_id	uuid	No	-	FK → report_revisions(id)
to_revision_id	uuid	No	-	FK → report_revisions(id)
previous_content	jsonb	No	-	Content from previous revision
new_content	jsonb	No	-	Content in current revision
created_at	timestamptz	No	now()	-

Indexes: - `idx_change_records_report_section` on (report_id, section_id)

notifications

Column	Type	Nullable	Default	Constraints
id	uuid	No	gen_random_uuid()	PRIMARY KEY
user_id	uuid	No	-	FK → users(id) ON DELETE CASCADE
type	notification_type	No	-	-
title	varchar(255)	No	-	-
message	text	No	-	-
link	varchar(500)	Yes	NULL	Deep link to relevant page
is_read	boolean	No	false	-
created_at	timestamptz	No	now()	-

Indexes: - `idx_notifications_user_unread` on (user_id, is_read, created_at DESC) — notification bell query

audit_log

Column	Type	Nullable	Default	Constraints
id	uuid	No	gen_random_uuid()	PRIMARY KEY
user_id	uuid	No	-	FK → users(id)
action	varchar(100)	No	-	e.g., 'report.submit', 'template.publish'
entity_type	varchar(50)	No	-	e.g., 'report', 'template', 'comment'
entity_id	uuid	No	-	ID of the affected entity
metadata	jsonb	Yes	NULL	Additional context
created_at	timestamptz	No	now()	-

Indexes: - `idx_audit_entity` on (entity_type, entity_id) — find all actions on an entity - `idx_audit_user` on (user_id, created_at DESC) — user's action history

Example entries:

```
| action              | entity_type | metadata                                      |
|---------------------|-------------|-----------------------------------------------|
| report.create       | report      | { "template_id": "...", "version": 1 }      |
| report.submit       | report      | { "revision": 1, "submitted_to": "fmt" }    |
| report.revision     | report      | { "requested_by": "...", "comments": 3 }    |
| template.publish    | template    | { "version": 2 }                            |
| comment.create      | comment     | { "report_id": "...", "section_id": "..." } |
| user.invite         | user        | { "organization_id": "...", "role": "MEMBER"}|
```

Relationship Summary

Parent	Child	Relationship	ON DELETE
organizations	users	1:N (organization_id)	RESTRICT
organizations	countries	1:1 (organization_id)	RESTRICT
organizations	countries	0:1 (auditor_organization_id)	SET NULL
users	report_templates	1:N (created_by)	RESTRICT
report_templates	template_sections	1:N (template_id)	CASCADE
report_templates	reports	1:N (template_id)	RESTRICT
countries	reports	1:N (country_id)	RESTRICT
users	reports	1:N (created_by)	RESTRICT
reports	report_sections	1:N (report_id)	CASCADE
reports	report_revisions	1:N (report_id)	CASCADE
reports	comments	1:N (report_id)	CASCADE
reports	change_records	1:N (report_id)	CASCADE
report_sections	comments	1:N (section_id)	CASCADE
report_revisions	comments	1:N (revision_id)	RESTRICT
comments	comments	1:N (parent_comment_id)	CASCADE
users	notifications	1:N (user_id)	CASCADE
users	audit_log	1:N (user_id)	RESTRICT

Seed Data (Development)

```
-- FMT Organization
INSERT INTO organizations (id, name, type) VALUES
  ('org-fmt-001', 'World Bank FMT', 'WORLD_BANK');

-- FMT Admin User (cognito_sub set after Cognito user creation)
INSERT INTO users (id, cognito_sub, email, first_name, last_name, organization_id, role) VALUES
  ('usr-admin-001', 'cognito-sub-placeholder', 'admin@worldbank.org', 'FMT', 'Admin', 'org-fmt-001', '

-- Sample Country Organizations
INSERT INTO organizations (id, name, type) VALUES
  ('org-eth-001', 'Ethiopia Climate Authority', 'COUNTRY'),
  ('org-col-001', 'Colombia Environmental Agency', 'COUNTRY');

-- Sample Countries
INSERT INTO countries (id, name, iso_code, organization_id) VALUES
  ('cty-eth-001', 'Ethiopia', 'ETH', 'org-eth-001'),
  ('cty-col-001', 'Colombia', 'COL', 'org-col-001');

-- Sample Auditor Organization
INSERT INTO organizations (id, name, type) VALUES
  ('org-aud-001', 'Global Audit Partners', 'AUDITOR');
```